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Sound Sight- A TTS and STT Integrated Model using OCR and ASR

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Abstract- This paper discusses the feasibility of using an automated pipeline that is text-to-speech and speech-to-text, which will be powered with the usage of Optical Character Recognition (OCR) that may be used for extracting text from images or documents and Automatic Speech Recognition (ASR) which can be used for the benefit of using algorithms and machine learning techniques to give more accurate results. The paper also explores the system architecture, with also a look at the methodology as it implements the technologies.

Keywords—Optical Character Recognition, Automatic Speech Recognition, Text-to-speech, Speech-to-text

I. INTRODUCTION

Perhaps most importantly, TTS and STT now have emerged as staples of modern human-computer interaction as voice-based interfaces grow increasingly ubiquitous in various applications, including virtual assistants, customer service automation, and accessibility tools to people with different disabilities. Advanced capabilities in AI and ML have been the drivers of significant improvements in accuracy, efficiency, and versatility for these systems. Yet, despite these advances, several critical challenges remain, hindering the widespread adoption and optimal performance of TTS and STT systems in real-world applications.

Some of the major issues these systems face are noise interference. In real-world environments, background noises, echoes, and overlapping speech have a degrading effect on the performance of STT systems, resulting in a problem in transcriptions and reliability. Another critical issue is accent and dialect variation. There are not enough developed models trained on the full range of global accents and regional dialects. Most current STT models are primarily trained only on a limited set of datasets that represent only regional varieties, and accuracy degrades when used by native speakers with a strong accent or against other variants of a language.

Another challenge for TTS systems is the naturalness of speech synthesis. Despite recent advances in the neural network model, such as WaveNet and Tacotron, which have yielded much better quality synthesized speech, unnatural prosody, an unnatural robotic-like tone, and lack of emotional expression still remain problems. The most important one would be that the speech output is human-like in intonation, rhythm, and emotion, if applications like a virtual assistant areanticipated.

To address these challenges, this paper introduces a new approach that combines the benefits of latest AI developments in the likes of ChatGPT integration with TTS and STT systems. The system will allow for increased speech-to-text conversion accuracy and naturalness during text-to-speech synthesis in general because it makes full use of transformer- based architectures and deep learning models.

Contributions of this work are two-fold: demonstrating the practical applications of state-of-the-art AI models such as ChatGPT and architectures based on transformers to boost TTS and STT, and proposing a solid system tackling interference noise, diversity in accents, and naturalness issues related to speech synthesis. These basic functionalities do focus, providing a scalable and flexible solution which would be deployable in a wide variety of real-world applications: virtual assistants, automated transcription services, and accessibility tools for the visually or hearing impaired.

II.RELATED WORK

Isewon et al. T, Akande T, Ganiyu R, and Okolie C (2024) [2] proposed the TTS text-to-speech system. This system was designed for a visually challenged citizen, where text was converted to speech. Thus, the system is essentially the most coveted tool for visually impaired citizens to gain access to written information. The algorithms which were developed that enabled speech output to be intelligible and clear. Even though the system has shown to be successful in helping the visually impaired, it might be somewhat lesser in terms of performance by the quality of the text input and the different resources which are required computational wise for processing in real time.

Liu, Y., Zhou, Z., Yang, W., & Liu, B. (2023) [4] also proposed a model which was specifically designed for the strong STT translation. The architecture created by them challenges noise and the difference noted in speech, hence ensuring the reliability and accuracy of transcriptions. The advancement in the architectures on top of the network, for example, transformer, upgrade the understanding capabilities and address different type of accents, dialects, and background noises. The authors were also able to show that there was some reasonable improvement in the performance of translation relative to traditional STT systems in challenging acoustic environments. Due to its high scalability and high efficiency, there is high promise to realize real- world applications in very diverse settings with this model.

R. Kumar and A. Singh, who developed a Noise Robust Training of the Deep Neural Networks for Improved Accuracy of Speech-to-Text in Noisy Environments, 2022 [5], where the authors presented a noise-robust training scheme which could be used to enhance the ability of the noise filtration process of the proposed model, and retain some amount of the speaker's voice. Significant relative improvements are shown over traditional STT systems, which can be seen in real-world noisy conditions. This research points out the possibility of the more sophisticated neural networks, which might address the issues present in the common STT technology.

N. K. Sawant & S. Borkar presented the " Development of Optical Character Recognition (OCR) Based System for Converting Devanagari Printed Text-to-Speech" where using OCR, the printed Devanagari text was converted to speech, by N. K. Sawant & S. Borkar in 2021. In addition, mostly, thesystem works on Marathi text; OCR recognizes the printed characters that are further processed and then converted into speech by TTS technology. The paper describes an important step forward for improving accessibility for the visually impaired: enabling automatic reading of printed Devanagari text. So, here it is: accessible systems can now become more engaging and user-friendly with accurate text recognition and natural-sounding clear speech output.

III. THEORETICAL BACKGROUND

A major breakthrough in developing architectures is achieved with the inclusion of Transformer-based models and ChatGPT in developing Speech-to-Text (STT) and Text-to- Speech (TTS) systems. This chapter studies the functionality of these models, as well as the role played by Optical Character Recognition in the text extraction process.

A. Transformer Models in Speech-to-Text (STT)

The transformer model, first proposed by Vaswani et al. in "Attention is All You Need," has become the spine of all types of modern STT systems. It is one of a kind where, a mechanism which allows the model to weigh-in how important are the different parts of the input data, where the position does not matter and it is really essential for speech processing because audio segments may have varying importance to the context.

Transformers are excellent at handling long-range dependencies in speech; these models capture the context words uttered earlier in a conversation with high accuracy and render better transcriptions. The transformer model BERT, along with its variation Wav2Vec 2.0, that uses large datasets of speech data from various environments can improve transcription quality up to an acceptable level, especially when dealing with diverse accents, regional dialects, or noisy backgrounds. These models also easily integrate with self-supervised learning, which allows them tocontinuously learn from unlabeled speech data and thereforeimprove their performance over time.

Hence the models can be improved, improvised and learnt from overtime and thus can be enhanced in such a way that the translations from various languages can be done at ease and also in an efficient manner.

B. Usage of ChatGPT in the field of Text-to-Speech (TTS)

A key role is assigned to ChatGPT: that is, it creates human-like text and transforms it into speech. Being a generative language model, ChatGPT uses GPT architecture in making contextually appropriate responses; by this, the text provided for TTS conversion would be coherent and contextually relevant. ChatGPT will also support the generation of new and different types of natural speech by creating linguistically correct and accurate user-input-specific response formulations.

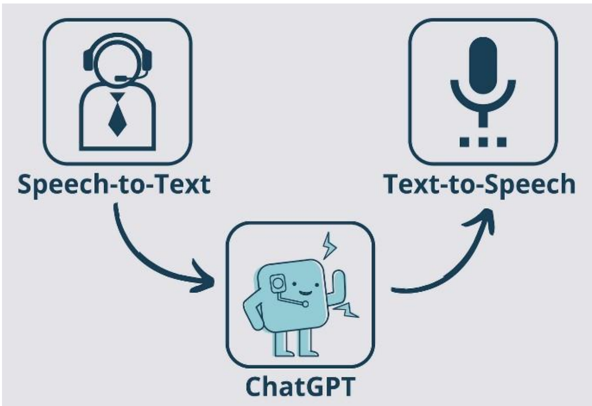


Fig.1. represents the use of ChatGPT in Text-to-speech process

C. Feature Extraction in OCR for TTS

For Optical Character Recognition (OCR), Feature extraction in this case is the greatest feature in character recognition of the text scanned. In this case, Histogram of Oriented Gradients is one of the most widely used object detection techniques to enhance character recognition in printed documents. HOG captures the edge features which are highly useful in differentiating between characters especially in complex scripts like Devanagari or Marathi.

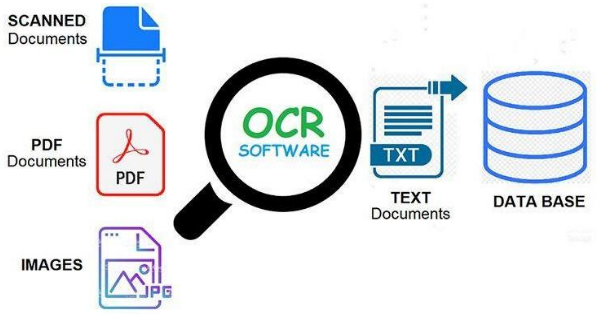


Fig.1. represents the use of OCR in Text-to-speech process

In these complex scripts the OCR software comes in very handy as the different characters and different styles can be picked up very easily and hence the process would be quite efficient aswell. Thus the decision was made to include the usage of OCR for its benefits as it would be helpful for this cause.

The processed text which is received from the process is thenready to be converted with the use of the TTS module.

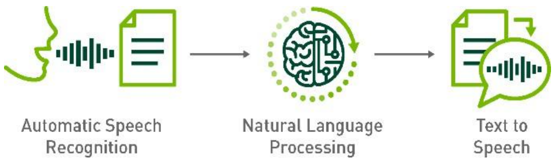


Fig.2. represents the Text-to-speech process


Fig.3. represents the Speech-to-text process

IV. Proposed Methodology

The proposed system will contain various integrated modules. One module for each of the varieties, especially in regards to distinctive features and models concerning speech processing and text processing. The most updated AI models accessible along with efficient techniques will be adopted within this system for correctness and efficiency of both STT and TTS operations. Given below is the outline of the key modules forming the system :

A. OCR Module

The OCR module transforms the image of text scanned from a variety of sources into an edit-friendly format and machine-readable text. The system uses Support Vector Machine (SVM) classifiers and Histogram of Oriented Gradients as techniques for feature extraction. Such a combination allows complex characters, such as those encountered in ordinary usage of the Devanagari script, while working with other regional scripts. After text extraction, the text now undergoes a preprocess stage to eliminate all noise, artifacts that would have been introduced during the scanning.

1. Image Preprocessing: This process is involved in converting the scanned document into a grayscale format and will then apply filtering to enhance the text and various other features.
2. Character Segmentation: The process in which the text issplit into individual characters where they can be identified, which can be then classified using the SVM classifier.
3. Text Conversion: The recognized characters will then be able to be converted into a type of editable text, which can then be further processed for different types of speech synthesis.

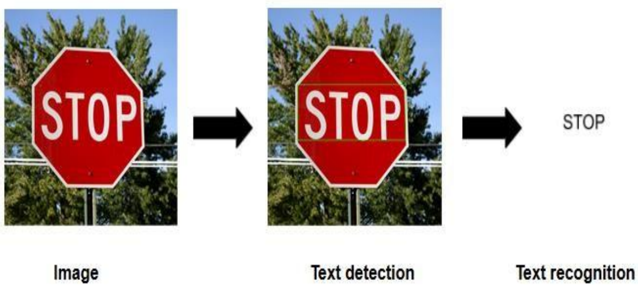


Fig.4. OCR process

B. ChatGPT Integration

The availability of ChatGPT gives the system conversational AI capabilities and ensures contextual, coherent responses. Building on GPT-4 architecture, ChatGPT handles NLP tasks and enables the system to create the right kind of response in line with user input, thereby producing extraordinary utility applications such as virtual assistants-for instance, where the user would expect fluid context-aware conversations.

1. Contextual understanding: It uses transformer-based architectures to understand the context that a given conversation will be aligned with so that the responses generated are tailored for the input posted by the user.
2. Dynamic Conversation: It will be dynamic for the queries from users, which return appropriate and accurate information on-the-fly.
3. STT and TTS compatibility: They are STT and TTS compatible because they can have spoken input translated to text, they can then generate a reply, and finally translate the reply back into speech for the

user.

4. Usage of OCR: The use of OCR would make sure that the methods mentioned above would be used properly and thus the results would be shown with more accuracy.

C. STT Module

The Speech-to-Text (STT) module will be used to transcribe the spoken language provided, into text usingthe various advanced speech recognition models. The modules leverage different deep learning techniques, which include the convolutional neural networks (CNNs) and recurrent neural networks (RNNs), to makesure that they accurately transcribe speech in real-time. The STT module is designed to be trained on large datasets, which would have different and diverse speech samples involved which makes it ready and useful for, enabling it to handle various accents, dialects, and noisyenvironments.

Input: The Input which would be used will be either Audio recordings, live speech input, text used from different paragraphs from books articles, etc.
Processing: The audio input is processed through a pipeline that includes pre- processing (noise reduction, normalization), feature extraction (MFCCs, spectrograms), and decoding (using HMM or neural network-based models).

D. ***TTS Module***

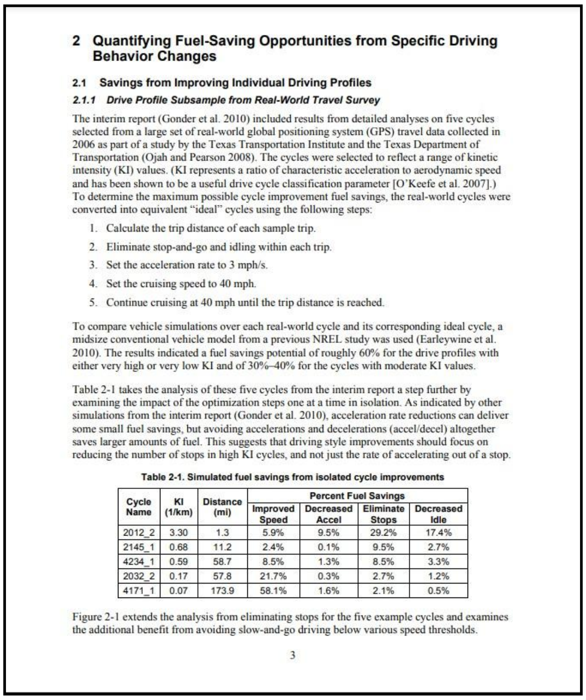
The Text-to-Speech (TTS) module converts text into natural-sounding speech. This module uses neural TTS models, such as Tacotron 2 and Wave Net, which produce high-quality speech with natural prosody and intonation. The TTS module ensures that the system can deliver responses in a human- like voice, enhancing the user experience by making interactions more lifelike and engaging.

Input: Text generated by getting the sample text from the files given by the receiver, which would be then used for conversion.
Processing: The input text is processed through a TTS pipeline that includes text normalization, phoneme conversion, prosody modeling, and waveform synthesis.

V. RESULT ANALYSIS

The system was successful in exhibiting the results for the Text-to-speech model, Speech-to-text model and the usage of Optical Character Recognition in the most efficient and accurate way possible. The process was carried out in different areas such as quite areas and also in noisy background areas. The OCR system was also tested by using different documents of various types and sizes and also involving the use of tables in the images used for OCR. For TTS module, various texts and paragraphs of different sizes and languages were used in order to check the efficiency of the module.

As for the execution process it was presented in the way where first the image for OCR conversion was first taken andpresented:



2 Quantifying Fuel-Saving Opportunities from Specific Driving Behavior Changes

2.1 Savings from Improving Individual Driving Profiles

2.1.1 Drive Profile Subsample from Real-World Travel Survey

The interim report (Gonder et al. 2010) included results from detailed analyses on five cycles selected from a large set of real-world global positioning system (GPS) travel data collected in 2006 as part of a study by the Texas Transportation Institute and the Texas Department of Transportation (Ojah and Pearson 2008). The cycles were selected to reflect a range of kinetic intensity (KI) values. (KI represents a ratio of characteristic acceleration to aerodynamic speed and has been shown to be a useful drive cycle classification parameter [O'Keefe et al. 2007].) To determine the maximum possible cycle improvement fuel savings, the real-world cycles were converted into equivalent "ideal" cycles using the following steps:

1. Calculate the trip distance of each sample trip.
2. Eliminate stop-and-go and idling within each trip.
3. Set the acceleration rate to 3 mph/s.
4. Set the cruising speed to 40 mph.
5. Continue cruising at 40 mph until the trip distance is reached.

To compare vehicle simulations over each real-world cycle and its corresponding ideal cycle, a midsize conventional vehicle model from a previous NREL study was used (Earleywine et al. 2010). The results indicated a fuel savings potential of roughly 60% for the drive profiles with either very high or very low KI and of 30%–40% for the cycles with moderate KI values.

Table 2-1 takes the analysis of these five cycles from the interim report a step further by examining the impact of the optimization steps one at a time in isolation. As indicated by other simulations from the interim report (Gonder et al. 2010), acceleration rate reductions can deliver some small fuel savings, but avoiding accelerations and decelerations (accel/decel) altogether saves larger amounts of fuel. This suggests that driving style improvements should focus on reducing the number of stops in high KI cycles, and not just the rate of accelerating out of a stop.

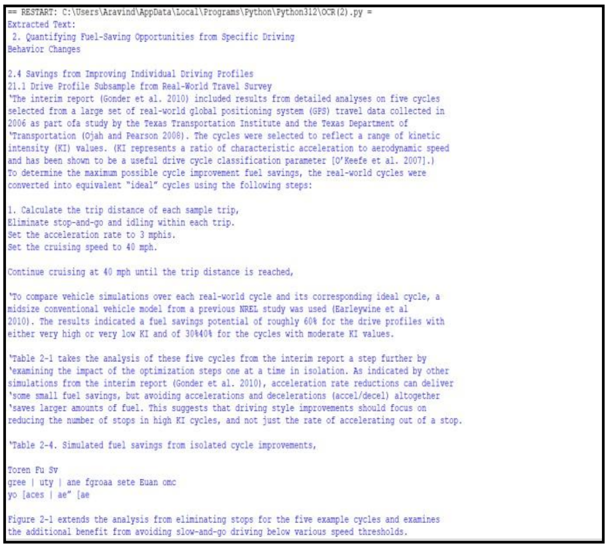
Table 2-1. Simulated fuel savings from isolated cycle improvements

Cycle Name	KI (1/km)	Distance (mi)	Percent Fuel Savings			
			Improved Speed	Decreased Accel	Eliminate Stops	Decreased Idle
2012_2	3.30	1.3	5.9%	9.5%	29.2%	17.4%
2145_1	0.68	11.2	2.4%	0.1%	9.5%	2.7%
4234_1	0.59	58.7	8.5%	1.3%	8.5%	3.3%
2032_2	0.17	57.8	21.7%	0.3%	2.7%	1.2%
4171_1	0.07	173.9	58.1%	1.6%	2.1%	0.5%

Figure 2-1 extends the analysis from eliminating stops for the five example cycles and examines the additional benefit from avoiding slow-and-go driving below various speed thresholds.

3

The above pdf discusses about the different fuel saving opportunities from specific driving behavior changes. Here we would take the OCR program to be executed and get the text and tables from the pdf with the use of different libraries which include: PymuPDF, Pdf2image, etc. For extracting the information present in the table, Camelot is used for getting the information and presenting it in the output format.



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== RESTART: C:\Users\Aravind\AppData\Local\Programs\Python\Python312\OCR(2).py =
Extracted Text:
 2. Quantifying Fuel-Saving Opportunities from Specific Driving
Behavior Changes

2.4 Savings from Improving Individual Driving Profiles
21.1 Drive Profile Subsample from Real-World Travel Survey
'The interim report (Gonder et al. 2010) included results from detailed analyses on five cycles
selected from a large set of real-world global positioning system (GPS) travel data collected in
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'Transportation (Ojah and Pearson 2008). The cycles were selected to reflect a range of kinetic
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'saves larger amounts of fuel. This suggests that driving style improvements should focus on
reducing the number of stops in high KI cycles, and not just the rate of accelerating out of a stop.

'Table 2-4. Simulated fuel savings from isolated cycle improvements,

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Figure 2-1 extends the analysis from eliminating stops for the five example cycles and examines
the additional benefit from avoiding slow-and-go driving below various speed thresholds.
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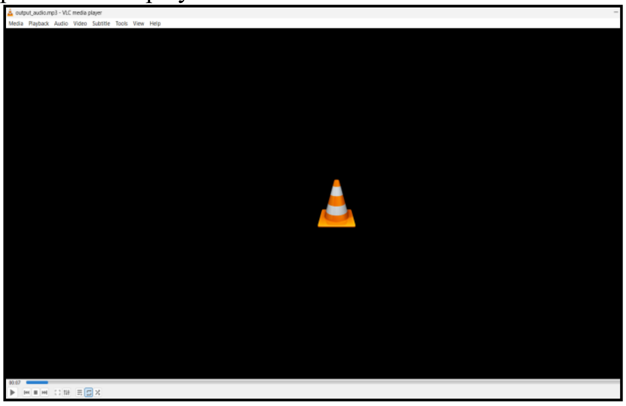
For the next part, we were able to execute the Text-to-Speech method where the required text was able to be extracted and converted to an audio file and be able to be played and listened to.

The extracted text would be converted into an audio file in the WAV format. The converted audio file would be saved in the default directory stated and can be viewed and listened to.

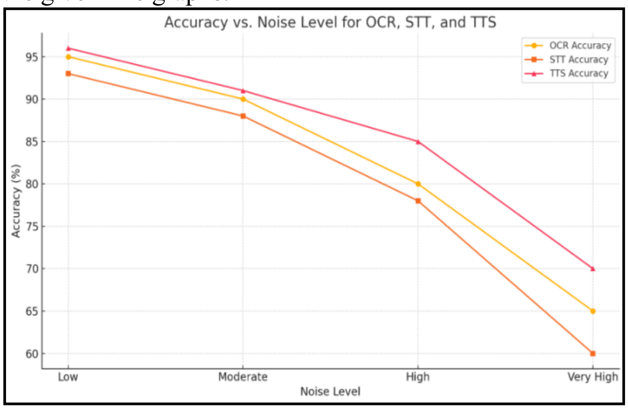
The audio file would be played with a voice being presentedand played for the whole duration.



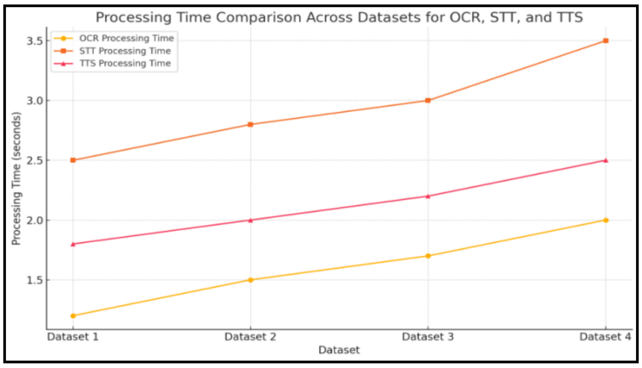
For the Speech-to-text module, the user can provide their recorded files to the module and then the process would takeplace where the text present in the audio file would be converted into a text file which can be seen and viewed by the user.

Thus all the above mentioned facilities are placed into a single menu where the user can access the menu and choose their option of choice.

The Performance analysis and results can be justified by the given line graphs:



Accuracy vs Noise Level Graph:
The graph shown above mentions the accuracy of OCR, TTS and STT systems decreases as the noise levels increases, demonstrating the system's to different type of noise variations which might be present in the different type of locations and surroundings. With taking into account the different scenarios and noise levels, the system's capabilities were tested and thus the results are shown.



Processing Time Comparison Across Datasets for OCR, STT and TTS:
The graph mentioned above compares the different times of the processing module, which includes OCR, STT and the TTS systems across different datasets, which mentions the efficiency levels of the system and the capabilities of the model.

VI. Conclusion

The proposed system combines OCR, STT, and TTS functionalities with conversational AI capabilities through ChatGPT. By leveraging state-of-the-art transformer models and neural networks, the system provides a robust solution for real-time speech and text processing, ensuring high accuracy, natural speech synthesis, and broad language support.

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